Provakar Mondol

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[Google Scholar] [LinkedIn]

Research Interest

- Deep Learning
- Image Processing
- Video Processing
- Medical Imaging
- Health Analytics
- Neuroscience

Education

- **Khulna University of Engineering and Technology (KUET)** — Khulna, Bangladesh
  M.Sc. *in Electrical and Electronic Engineering; CGPA: 3.83/4.00* — July 2021 – Ongoing
  Thesis: *Optimized Deep Learning Models for Gastro-intestinal Image Classification and Automated Polyp Detection*
- **Khulna University of Engineering and Technology (KUET)** — Khulna, Bangladesh
  B.Sc. *in Electrical and Electronic Engineering; CGPA: 3.52/4.00* — March 2013 – May 2017
  Thesis: *Design of A New Dry Electrode and Comparison of Performance with Biopac Wet Electrode*

Research Experience

- **Gastrointestinal Image Classification and Automated Polyp Detection (2023-Present)**
  - We **classified the multiclass GI image data** using customized CNN to aid the endoscopic and colonoscopic inspection.
  - Performed **panoptic segmentation on Polyp image** data using customized U-Net and Watershed algorithm to better understand the polyp condition in colon.
  - Working on development of a **real-time polyp detection system** from colonoscopic videos to provide an automated and time-saving solution to the problem.
- **Framework for Diagnosis of Alzheimer's Disease in Data-Constrained Scenarios (2023-Present)**
  - Compared **performances of augmentation and SMOTE on unbalanced MRI image data for classification** of stages of Alzheimer's Disease using a customized deep learning model.
  - Applied **transfer learning using three state-of-the-art models** to evaluate their performance on classification.
  - Working on **generating synthetic image data using WGAN-GP** to further solve data unbalance constraints.
- **Development of A Dry Electrode for Bio Signal Acquisition (2016-2017)**
  - **Designed and developed a new type of dry electrode** which can extract bio signals like ECG, EEG and EMG.
  - Compared the **performance of signal acquisition** using developed dry electrode with those acquired using state-of-the-art Biopac wet electrodes.
- **Short Channel Effects Suppression in a Dual-Gate Gate-All-Around Si Nanowire Devices (2015-2016)**
  - Designed a Dual-Gate GAA Silicon nanowire nMOSFET and extracted its short channel parameters and compared the parameters with those of GAA.
  - Studied the process sensitivity and mobility degradation of the designed nMOSFET.

Publications

- **P. Mondol**, P. Das, and K. K. Halder, "U-Net and Watershed Based Deep Learning Approach for Panoptic Segmentation of Colorectal Polyps in Colonoscopy," IEEE International Conference on Electrical, Computer and Communication Engineering (ECCE), Chittagong, Bangladesh, 2025.
- **P. Mondol**, P. Das, and K. K. Halder, "GINet: A Deep Learning Approach to Gastrointestinal Image Classification," IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh, 2025.

- P. Das, M. Islam, and **P. Mondol**, "Utilizing a Multi-Class CNN Model for Precise Diagnosis of Alzheimer's Disease," IEEE Global Conference on Cognitive Computing and Communication Technology (GC4T), Pune, India, 2025.
- P. Das, M. Islam, and **P. Mondol**, "A Comparative Analysis of Pre-trained Models for Identifying Alzheimer's Disease," IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh, 2025.
- M. W. Rony, **P. Mondol**, and H. R. Myler, "Sensitivity of a 10nm dual-gate GAA Si Nanowire nMOSFET to Process Variation," 19th International Conference on Computer and Information Technology (ICCIT), Dhaka, Bangladesh, 2016.
- M. W. Rony, P. Bhowmik, H. R. Myler, and **P. Mondol**, "Short Channel Effects Suppression in a Dual-gate Gate-All-Around Si Nanowire Junctionless nMOSFET," 9th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, 2016.

PROFESSIONAL EXPERIENCES

- **Bangladesh India Friendship Power Company Limited** — Bagerhat, Bangladesh
  Deputy Manager, Department of Control & Instrumentation — December 2020 – Present
  Responsibilities: Open-loop, closed-loop and coordinated master control loop tuning and healthiness checking. Implementation and modification of Graphical User Interface of the system.
- **Daffodil International University (DIU)** — Dhaka, Bangladesh
  Lecturer, Department of EEE — October 2017 – November 2020
  Responsibilities: Teaching undergraduate students on Control Systems, Electrical Machine, Electronic Devices and Circuit Theory, Analog Electronics, Digital Electronics, Random Signals and Processes, Power System Protection, Microprocessors and Interfacing along with labs of these courses.
- **Prime University (DIU)** — Dhaka, Bangladesh
  Lecturer, Department of EEE — August 2017 – September 2017
  Responsibilities: Teaching undergraduate students on Control Systems, Electrical Machine, Electronic Devices and Circuit Theory.

TECHNICAL & PERSONAL SKILLS

- **Programming Languages**: Python, C, C++, MATLAB
- **Frameworks**: Keras, TensorFlow
- **Hardware & Circuit Design**: Arduino, Atmel AVR, Proteus
- **Graphics Programming and Industrial Applications:** Siemens SPPA T3000, MaxDNA, Valmet VMS, Honeywell PLC, Schneider PLC, Emerson PLC, ABB PLC

HONORS AND AWARDS (SELECTED)

- **Dean's Award** in fourth year in KUET for outstanding academic results
- **Board Scholarship (Merit-13)** in Rajshahi Board for outstanding academic results in Higher Secondary School Certificate Exam-2012
- **Board Scholarship (Merit-41)** in Rajshahi Board for outstanding academic results in Secondary School Certificate Exam-2010

SUPERVISION EXPERIENCES (SELECTED)

- Supervised in a Line-Follower robot competition as a mentor. My overall role was to maintain the rules of the competition and give marks on competitors' robot performance. (December 2016)
- Worked as a technical core and event promotional member in the Inter-University Tech Fiesta (IUTF), Khulna University of Engineering & Technology (KUET), a nationwide science and technology competition and research platform. (January 2016)
- Volunteered in the International Conference on Electrical Information and Communication Technology (EICT), Khulna, Bangladesh as a technical operator. (December 2015)

Language Proficiency

- **Bengali** (Bangladesh, Native)
- **English** (Fluent)
- **Hindi** (Fluent)

References

- **Dr. Kalyan Kumar Halder**
  Professor, Department of EEE
  Khulna University of Engineering and Technology (KUET), Khulna, Bangladesh
  Email: kalyan@eee.kuet.ac.bd
  [Google Scholar]

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